

# **EduPath AI - Major Recommendation System**

## **Bias Analyst Report**

### **1. Sensitive Feature Identification**

Out of the input parameters utilized as Machine Learning inputs according to the system poster (Academic Scores, Interests & Achievements, and RIASEC Assessment), several sensitive and structural features possess the potential to introduce algorithmic bias if not properly addressed. These features include:

- Target Class (Major Label): The recommended major column indicating the student's ideal academic path (30 unique majors).
- RIASEC Assessment Profiles: Columns representing Holland personality types (R, I, A, S, E, C) which may correlate with socio-cultural gender roles or environmental socialization.
- Interests & Achievements: The achievement level (prestasi) and achievement field (bidang\_prestasi) parameters, which often act as proxies for socioeconomic class and resource availability.
- Synthetic Generation Bias: The artificial mathematical rules (Gaussian noise around ideal centers) that separate major profiles too cleanly without representing realistic, cross-disciplinary, or borderline students.

### **2. Descriptive Statistics**

Based on direct extraction from the synthetic training dataset of 3,000 student records, the distribution of features exhibits the following characteristics:

- Class Balance (Perfect Uniformity): The dataset is perfectly balanced with exactly 100 records (3.33%) for each of the 30 majors. While computationally convenient, this is highly unrepresentative of real-world major distribution, where certain fields experience massive demand while others are minority classes.
- Academic Scores Distribution: The 8 academic subjects are generated using normal distributions centered around ideal profiles (ranging from 30.0 for low-relevance to 90.0 for high-relevance). This creates artificial clusters with no overlapping outliers, representing a population of "idealized" students.
- RIASEC Assessment Distribution: The 6 RIASEC dimensions are centered at discrete interest levels (scale 6 to 30), creating a highly structured personality profile that does not account for students with flat profiles (high in all or low in all dimensions).
- Achievement Distribution: Extracurricular achievements are added as discrete levels. Because the dataset is synthetic, it assumes a uniform distribution of achievements across majors, which does not reflect the real-world disparity where resource-rich environments yield significantly higher achievement counts.

### **3. Impact Analysis**

If these imbalances and sensitive features are left unmitigated during AI training, the consequences could be detrimental in educational counseling applications:

- **Impact of Artificial Class Balance:** Since the model is trained on a perfectly uniform distribution of 100 samples per major, it will fail to adapt to real-world base rate probabilities. It might over-recommend low-demand majors and under-recommend high-demand majors compared to actual university capacity and job market realities.
- **Impact of Achievements & Socioeconomics:** Since achievements weight the Euclidean distance, students from wealthier families who have greater access to private tutoring, sports clubs, and science olympiads will receive artificial similarity boosts to competitive majors, while equally capable under-resourced students are algorithmically penalized.
- **Impact of RIASEC Gender Bias:** If the RIASEC interest scoring maps directly to historical gender major distributions (e.g., socialized interests pointing females to humanities/social sciences and males to STEM), the AI will reinforce and lock in these traditional gender imbalances, acting as a barrier to diversity in fields like Engineering.
- **Impact of Synthetic Bias:** The perfectly drawn boundaries (Gaussian noise with  $SD=8.0$  for academics and  $SD=3.0$  for RIASEC) cause the KNN classifier to yield absolute certainty when matching a student. In a real-world setting where students have mixed, ambiguous, or multi-disciplinary strengths, this overconfidence could pigeonhole students into narrow, inappropriate fields.

#### 4. Mitigation Actions

To neutralize these biases and establish a fair Educational Decision Support System (Algorithmic Fairness) in accordance with the poster's guidelines, this system implements the following mitigation strategies:

1. KD-Tree Similarity Matching with
2.  $K=5$
3.  $K=5$ : Utilizing a K-Nearest Neighbors (KNN) algorithm with
4.  $K=5$
5.  $K=5$  and a KD-Tree structure. By retrieving similarity matches from only the closest 5 metric profiles, the model prevents remote, irrelevant data samples from voting, ensuring a localized and highly accurate recommendation.
6. **Objective Feature Isolation (Demographic Fairness):** Excluding explicit demographic features (such as gender, age, religion, or family income) from the input vector. The model performs classification purely on normalized Academic Scores, Interests & Achievements, and RIASEC Assessment scores to maintain demographic parity.
7. **Explainable AI & Counselor Safety Net (Transparency & Human Oversight):** Every recommendation output is accompanied by a transparent natural-language explanation generated by the backend. Rather than operating as an absolute "black-box" decision maker, the system presents recommendations as suggestions and includes a mandatory prompt advising students to consult a human school counselor before making final educational choices.
8. **System Limitations (Final Evaluation Disclaimer):** It is documented as a key limitation that the model is trained on synthetic data. Before deploying this application in a school district or university, the KNN engine must be retrained and calibrated using a real-world, anonymized database of historical student performance and successful major alignment to dissolve the synthetic Gaussian boundary bias.

